

Leighton Relationship Code and Scientific Validity Review

Repository: /Volumes/External_Drive/Research/Leighton Relationship

Audit date: 28 July 2026

Review type: Read-only, adversarial correctness and reproducibility assessment

Overall assessment

The current pipeline is not yet suitable for defensible scientific interpretation. The conventional Leighton-ratio equation is present, but the working notebook mixes monitoring sites, the AQS ingestion request is invalid, almost all analyzed NO is below the recorded detection limit, and timestamp and calibration choices can change the result materially.

Prepared as a durable record of the code review. No repository source files were modified by the audit.

Executive summary

The primary problem is not the algebraic form of the Leighton relationship. It is the provenance and treatment of the quantities inserted into it. The current working notebook loads Lindon chemistry, pairs it with Hawthorne UV, writes a Hawthorne-named output, and interprets a ratio dominated by NO values below the monitor's recorded method detection limit. At the same time, the scraper requests more parameter codes than the EPA API permits and collapses monitor-level records without preserving POC, method, or qualifier information.

A targeted sensitivity analysis showed that the median ratio of 1.122 falls to 0.300 when NO is illustratively floored at half the recorded detection limit and to 0.150 at the full limit. These are not proposed corrections; they demonstrate that the apparent proximity to unity is not robust to the unresolved censoring policy.

Highest-priority conclusions

- Do not interpret or publish the current ratio until chemistry, UV, and meteorology are tied to one documented site and time basis.
- Repair AQS ingestion before collecting more data: split parameter requests, validate API-level status, preserve long-form records, and retain monitor provenance.
- Recover the UV/JNO2 instrument documentation and calibration equation. The local formula change multiplies every result by 3.521.
- Treat below-detection NO and NO2 explicitly and propagate their uncertainty; a positivity filter is not an adequate censoring model.
- Move the notebook's hidden choices into a parameterized, tested pipeline with immutable input/output metadata.

Severity overview

Severity	Count	Meaning
Critical	3	Current results invalid or core ingestion nonfunctional
High	5	Material scientific bias, data loss, or irreproducibility
Medium	6	Analysis, reporting, security, and validation weaknesses

Scope and method

The review traced the current working tree from data acquisition through the Leighton-ratio calculation and downstream exploration. It included the committed files, uncommitted notebook changes, the untracked LR test notebook, and the SR/clearing-index helper and tests that appeared while the audit was in progress.

- Inspected README, scrapers, notebooks, metadata, stored datasets, and tests.
- Compared working notebook source with the committed version to identify local regressions.
- Verified API constraints and time definitions against EPA and Synoptic documentation.
- Recomputed the current May 2025 selection and targeted sensitivity diagnostics.
- Ran available unit tests, static notebook syntax checks, git diff validation, and a mocked mixed-timezone Synoptic response.

Snapshot limitation: The repository was actively changing during the review. The main notebook and ignored output artifact changed while evidence was being collected. Findings refer to the latest source state observed on 28 July 2026, with local-regression status called out where known.

Detailed findings

CRITICAL

1. The working notebook combines measurements from different monitoring sites

The current Leighton-ratio result is physically uninterpretable because the chemical and photolysis inputs are not colocated.

- The main chemistry dataframe is loaded from data/Lindon/Lindon_raw.csv.
- The UV dataframe is restricted to the HW column, which denotes Hawthorne.
- The calculated result is written to data/Hawthorne/Leighton_Relationship_HW.csv.
- The committed notebook used Hawthorne chemistry; changing only the chemistry input to Lindon introduced this regression.

Impact: Any apparent agreement with unity can be an artifact of combining unrelated air masses, emissions, cloud conditions, and instrument histories.

Recommended action: Choose one site, select its matching UV or measured JNO2 series, assert site identity in code, and encode site metadata into the output filename and file contents.

Leighton_Relationship_Calculator.ipynb JSON lines 140, 145, and 315.

CRITICAL

2. The current AQS sample request violates the EPA parameter limit

The primary scraper requests six parameter codes even though the EPA AQS API permits at most five per request.

- PARAMS contains O3, NO, NO2, solar radiation, temperature, and UV.
- All six are concatenated into one sampleData/bySite request.
- The code accepts any HTTP 200 response without checking the AQS Header status or errors.
- The monitors/bySite discovery request omits its documented required param argument.

Impact: Site discovery may return no sites and data retrieval may silently become an empty dataset, making the pipeline nonfunctional or incomplete without a clear failure.

Recommended action: Split retrieval into requests of five or fewer parameters, validate both HTTP and AQS response status, surface API errors, and add a contract test using a recorded response.

Scrapers/scraper.py lines 41-49, 126, 155-168.

CRITICAL

3. The result is dominated by measurements below recorded detection limits

A simple NO = 0 filter is not an adequate treatment of censored or below-detection observations when NO is in the denominator.

- Lindon metadata records a 5.0 ppb NO detection limit and a 2.7 ppb NO2 detection limit.
- 122 of 123 retained NO observations are below 5.0 ppb; the minimum is 0.2 ppb.
- 45 of 123 retained NO2 observations are below 2.7 ppb.
- Illustrative NO floors changed the median ratio from 1.122 to 0.300 at 2.5 ppb and 0.150 at 5.0 ppb.

Impact: The conclusion that the median ratio is near unity is not robust to the unresolved censoring and reporting policy.

Recommended action: Recover the monitor-specific below-MDL policy, retain per-row detection limits, use an appropriate censored-data or uncertainty analysis, and report sensitivity bounds rather than a single ratio.

Leighton_Relationship_Calculator.ipynb lines 154-157; data/Lindon/Lindon_metadata.json lines 146-157 and 267-278.

HIGH

4. AQS and Synoptic timestamps use incompatible local-time conventions

The merger can be systematically one hour wrong during daylight-saving months and can fail on a range spanning both offsets.

- EPA AQS time_local is Local Standard Time; in Utah this is MST throughout the year.
- Synoptic is requested with obtimezone=local, which follows America/Denver and uses MDT in summer.

- The Synoptic code removes timezone information before merging to naive AQS timestamps.
- A mocked response spanning MDT and MST reproduced a mixed-timezone parsing failure and returned an empty dataframe.
- Stored datasets contain duplicated fallback-hour timestamps: 5 in Lindon and 70 in Hawthorne.

Impact: Solar-radiation values may be attached to the wrong chemical hour. Since radiation changes rapidly around morning and evening, this can alter filtering and any SR-to-UV calibration.

Recommended action: Use UTC internally for both sources, derive AQS UTC from `date_gmt/time_gmt`, retain timezone-aware timestamps, define hourly labeling explicitly, and assert timestamp uniqueness before merging.

Scrapers/SynopticScraper.py lines 87-122 and 129-137; Scrapers/scrapper.py line 269.

HIGH

5. Pivoting erases monitor provenance and quality flags

The object called raw data is already an averaged, lossy product.

- `pivot_table` averages all `sample_measurement` values sharing a timestamp and parameter.
- POC, method code, method type, detection limit, uncertainty, and qualifiers are not retained in the time series.
- Hawthorne metadata includes multiple NO and NO2 POCs and multiple historical detection limits.
- Individual calibration, malfunction, audit, fire, and voiding qualifiers cannot be reconstructed after pivoting.

Impact: Different instruments or qualified observations can be blended into a plausible-looking value, and later QA filtering becomes impossible.

Recommended action: Persist the API response as immutable long-form raw data. Select an explicit POC/method policy and row-level QA rules before creating a wide analysis table.

Scrapers/scrapper.py lines 272-298.

HIGH

6. Failure paths and local date-range changes can overwrite valid data

Both scrapers write directly over their inputs or canonical outputs without transactional protection.

- A missing Synoptic station writes an all-null `SR_Synoptic` column back to the input CSV.
- A parsing or network error returns an empty dataframe, which can also erase an existing Synoptic column.
- CSV and metadata writes are separate and non-atomic.
- The local `START_YEAR` change from 1980 to 2024 rewrites the same raw and cleaned filenames with a shortened history.

Impact: Transient failures or an exploratory date-range edit can silently destroy previously collected historical data.

Recommended action: Write to a new temporary file, validate row counts, coverage, uniqueness, units, and metadata, then atomically promote it. Make date bounds required CLI parameters and never overwrite raw inputs.

Scrapers/SynopticScraper.py lines 152-158 and 229-273; Scrapers/scrapper.py lines 230-326.

HIGH

7. The JNO2 calibration is undocumented and changes every ratio by 3.521x

The working calculation changed from $J = C \times UV$ to $J = C \times UV / 2.84 \times 10$ without a traceable derivation.

- The sensor area is hard-coded as 2.84 with no unit.
- The factor of ten has no explanation.
- The UV input file has no embedded unit or calibration metadata.
- `LR_test.ipynb` divides a quantity already labeled `J_NO2` by 2.84, creating a second, inconsistent convention.

Impact: Since JNO2 is in the numerator, the formula change multiplies the Leighton ratio by $10/2.84 = 3.521$ and can move conclusions across the unity reference line.

Recommended action: Obtain the instrument model, output units, spectral response, calibration certificate, and the exact conversion to s^{-1} . Encode units and provenance beside the calculation and test against measured JNO2.

Leighton_Relationship_Calculator.ipynb lines 223-225; LR_test.ipynb lines 26-28.

HIGH

8. Broad dropna and hidden filters create selection bias

Rows are discarded based on variables that do not enter the ratio even when the optional SR/CI filter is disabled.

- dropna is applied to the entire merged dataframe, implicitly requiring AQS SR and SR_Synoptic.
- The ratio itself uses UV, temperature, O3, NO, and NO2.
- UV > 10, positive gas filters, May 2025, and a fixed year are hard-coded rather than declared analysis parameters.
- Excluded-row counts and reasons are not recorded.

Impact: The result may represent unusually clear, fully instrumented hours rather than the target population, and the disabled filter does not restore an unfiltered sample.

Recommended action: Define a minimal required-column list for each analysis stage, parameterize every threshold, log a row-accounting table, and pre-register selection rules before inspecting the outcome.

Leighton_Relationship_Calculator.ipynb lines 154-170.

MEDIUM

9. Daytime and nighttime groups overlap and are physically mislabeled

The current interval definitions double-count boundary observations and the retained nighttime group is not actually dark.

- between_time is inclusive at both ends, so 09:00 and 16:00 appear in both groups.
- The diagnostic found 110 daytime rows, 39 nighttime rows, and 26 rows in both groups out of 123 unique rows.
- All retained observations already satisfy UV > 10, so the complementary clock interval cannot be interpreted as nighttime.

Impact: Histograms and medians are not independent comparisons and their labels overstate the physical interpretation.

Recommended action: Use non-overlapping intervals and classify daylight using solar elevation or measured JNO2 rather than clock time. Report group sizes and boundary policy.

Leighton_Relationship_Calculator.ipynb lines 321-322.

MEDIUM

10. Hourly averages are inserted into a nonlinear photostationary relationship

The conventional equation is correctly shaped, but its application assumes simultaneous measurements and a defensible photostationary-state timescale.

- The scraper creates hourly wide-form means before the ratio is calculated.
- A ratio of hourly means is not generally the mean of the instantaneous ratio.
- The nonlinearity is especially important when NO approaches zero or changes rapidly near emissions.
- Transport, peroxy-radical chemistry, and non-steady-state conditions can also drive departure from unity.

Impact: A value different from one cannot be uniquely attributed to chemistry, and a value near one may result from temporal aggregation.

Recommended action: Use the highest common time resolution available, test averaging sensitivity, estimate chemical timescales, and state the conditions under which the PSS interpretation is considered valid.

Scrapers/scraper.py line 298; Leighton_Relationship_Calculator.ipynb line 307.

MEDIUM

11. Pressure and uncertainty treatment are incomplete

The number-density conversion uses a fixed pressure and the notebook's uncertainty section does not propagate uncertainty.

- Pressure is fixed at 87,000 Pa for every site and hour.
- The result depends on number density and therefore on pressure.
- f_T is computed and plotted but not used in the ratio or an uncertainty model.
- No uncertainty is propagated from JNO2, NO, NO2, O3, temperature, pressure, censoring, calibration, or synchronization.

Impact: Reported medians and percentiles imply more precision than the inputs support and are not comparable across sites without a common pressure convention.

Recommended action: Use observed or elevation-adjusted pressure, document the approximation, and propagate uncertainty with analytical derivatives or Monte Carlo sampling.

Leighton_Relationship_Calculator.ipynb lines 201-205 and 246-259.

MEDIUM

12. The notebooks and stored artifacts are not reproducible

Several cells are stale, contradictory, path-dependent, or guaranteed to produce empty selections.

- Data_Explore.ipynb points to an unavailable /Volumes/T7 Touch path.
- It prints data.head rather than calling data.head().
- It intersects 2021-2023 good dates with a 2025 artifact and produces an empty dataframe.
- summer_2024 selects 2023 data after the dataframe has already been restricted to May 2025.
- The ignored output artifact changed during the review and did not match the notebook's displayed configuration.

Impact: Displayed output can be mistaken for a result produced by the current source, and another researcher cannot reproduce the analysis from a clean clone.

Recommended action: Clear stale notebook outputs, use repository-relative paths, execute from a clean environment, save configuration and input hashes with every artifact, and add an end-to-end smoke test.

Data_Explore.ipynb lines 71-73 and 299-320; Leighton_Relationship_Calculator.ipynb lines 437-439.

MEDIUM

13. Credential handling, TLS settings, and dependency documentation are unsafe

The scraper embeds fallback credentials and disables transport verification while the repository lacks a reproducible dependency declaration.

- A personal-looking email and API key are committed in scraper.py; the key is intentionally not reproduced in this report.
- EPA requests pass verify=False and global warnings are suppressed.
- The README states credentials are confined to an untracked .env, which is inaccurate.
- The repository .venv contains pandas but lacks requests and python-dotenv, so both scrapers fail to import there.

Impact: Credentials should be treated as exposed, TLS failures are hidden, and a clean environment cannot run the collection pipeline.

Recommended action: Rotate any real credential, remove fallbacks, require environment variables, restore certificate verification, narrow warning filters, and add a pinned dependency manifest.

Scrapers/scraper.py lines 13-35 and 75-163; README.MD line 25.

MEDIUM

14. The new SR/clearing-index filter has limited validation and ambiguous edge behavior

The helper's two unit tests pass, but important parsing, timezone, and physical-plausibility cases remain untested.

- The main notebook currently configures APPLY_SR_CI_FILTERS = False, so the helper is not active in the displayed source.
- $SR \geq 710 \text{ W/m}^2$ is accepted without a daylight or solar-position plausibility check.
- A 1000+ clearing-index token is converted to 1000 and retained by the ≤ 1000 rule, potentially losing the greater-than meaning.
- Tests do not cover archive parsing, mixed timezones, duplicate dates, 1000+, missing CI, HTTP failure, or notebook integration.
- The related Improved Isopleths source contains a verified midnight 1000 W/m^2 record, demonstrating that numeric thresholding alone cannot establish physical validity.

Impact: Invalid or misaligned radiation observations can survive the selection and make a clear-sky subset appear scientifically valid.

Recommended action: Add a daylight/solar-elevation guard, preserve qualified tokens such as 1000+, test recorded AFOS products and failure cases, and record exactly which filter version produced each output.

`sr_ci_filter.py` lines 120-122, 209-293; `tests/test_sr_ci_filter.py`.

Quantitative diagnostics

The following values were recomputed directly from the current Lindon chemistry and Hawthorne UV selection to quantify sensitivity. They describe the code as written, not a scientifically approved analysis dataset.

Diagnostic	Result	Interpretation
Rows in current May 2025 selection	123	After exact timestamp merge, positivity filters, dropna, and date filter
NO observations below 5 ppb MDL	122 / 123	99.2% of analyzed observations
NO2 observations below 2.7 ppb MDL	45 / 123	36.6% of analyzed observations
Minimum retained NO	0.2 ppb	NO appears in the denominator
Median ratio, current calculation	1.122	Before optional SR/CI filtering
Median with illustrative NO floor at 2.5 ppb	0.300	Sensitivity only; not a proposed correction
Median with illustrative NO floor at 5 ppb	0.150	Sensitivity only; not a proposed correction
Daytime rows	110	09:00 through 16:00, inclusive
Nighttime rows	39	16:00 through 09:00, inclusive
Rows counted in both groups	26	Boundary hours 09:00 and 16:00
J and ratio multiplier from local formula change	3.521x	10 / 2.84 relative to committed C x UV calculation

Stored-data integrity observations

- Lindon raw: 113,333 rows covering 2013-01-01 through 2025-12-31; 5 duplicate timestamps.
- Hawthorne raw: 249,695 rows covering 1997-01-01 through 2025-11-30; 70 duplicate timestamps.
- Smithfield and Vernal each contain 5 duplicate timestamps.
- Duplicate timestamps occur at the 01:00 daylight-saving fallback hour and are repeated identically in the current wide files.
- Existing files contain large missing-data blocks; complete-case filtering should therefore be audited by variable and year.

Verification results

Status	Check	Result
PASS	2 SR/CI unit tests	Both available tests completed successfully.
PASS	Notebook syntax	All code cells parsed under Python 3.12.
PASS	Whitespace validation	git diff --check reported no errors.

Status	Check	Result
FAIL	Mixed-offset Synoptic response	Mocked MDT/MST timestamps produced an empty return after parsing failure.
FAIL	Repository scraper environment	requests and python-dotenv are absent from the repository .venv.
NOT RUN	Live API collection	Avoided mutating external state and overwriting local datasets during a read-only review.
NOT RUN	End-to-end notebook execution	The main notebook writes over an ignored canonical output file.

Recommended remediation roadmap

Phase 1 - Stop invalid outputs and preserve evidence

1. Select one analysis site and immediately correct the chemistry/UV/output naming mismatch.
2. Rotate the committed AQS credential if real and remove all credential fallbacks.
3. Make all outputs append-only or content-addressed until validation passes.
4. Preserve current API responses and wide files as dated snapshots; do not silently rewrite them.

Phase 2 - Repair ingestion and time alignment

5. Split AQS parameter requests into valid batches and validate the AQS Header status.
6. Store immutable long-form observations with POC, method, units, MDL, uncertainty, qualifiers, date/time local, and date/time GMT.
7. Use UTC internally; resample with an explicit closed/label convention and validate one-to-one merges.
8. Choose monitor/POC and qualifier policies explicitly, then generate the analysis-wide table.

Phase 3 - Establish scientific measurement contracts

9. Document the UV sensor, units, collection geometry, calibration constant, area correction, factor of ten, and conversion to JNO2 in s^{-1} .
10. Reconcile the main and LR_test JNO2 conventions against an independent measured-J dataset.
11. Define below-detection treatment and propagate monitor-specific uncertainty.
12. Use observed pressure where available and quantify sensitivity to the fixed-pressure approximation.

Phase 4 - Validate the analysis

13. Compare instantaneous or higher-resolution ratios with ratios of hourly means.
14. Use solar elevation or measured photolysis to define daylight and make groups non-overlapping.
15. Add row-accounting, units, ranges, uniqueness, site identity, and timestamp assertions.
16. Create an end-to-end test with a small frozen dataset and a known expected ratio.
17. Only interpret departure from unity after documenting PSS, transport, and radical-chemistry limitations.

Suggested release gate

Do not use the output for scientific conclusions until all Critical findings are closed, the site/JNO2/MDL contracts are documented, an end-to-end frozen-data test passes, and a reviewer can reproduce the same artifact from a clean environment.

Technical references

[US EPA AQS API documentation](#)

Request limits, service inputs, response structure, and sample-data behavior.

[US EPA AirData file format documentation](#)

Local Standard Time definition, method detection limits, qualifiers, and units.

[US EPA AQCSV format](#)

POC and qualifier definitions.

[Synoptic Time Series API](#)

obtimezone behavior and time-series response structure.

[Carpenter et al., Atmospheric Chemistry and Physics \(2022\)](#)

Conventional Leighton ratio and photostationary-state interpretation.

Repository evidence index

Leighton_Relationship_Calculator.ipynb - Site selection, filtering, J, number-density conversion, ratio, grouping, and outputs.

Scrapers/scrapper.py - AQS API requests, monitor discovery, pivoting, cleaning, metadata, and overwrite behavior.

Scrapers/SynopticScrapper.py - Station selection, time parsing, hourly aggregation, merging, and in-place writes.

data/Lindon/Lindon_metadata.json - Units, POCs, methods, and detection limits.

LR_test.ipynb - Independent JNO2 test convention.

Data_Explore.ipynb - Downstream statistics, paths, and stale date filtering.

sr_ci_filter.py and **tests/test_sr_ci_filter.py** - Optional SR/clearing-index filter and current unit coverage.

End of report